

CHEAP 5G: DECT NR+ for Tactile Internet With Human In The Loop Application

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Abstract—5G communication networks are suitable to realize Tactile Internet with Human in the Loop application due to the envisaged latency promises. Currently, 5G Release 15 is widely deployed. Latency performance to control robots or enable humans to engage with the metaverse will require higher releases. WiFi is widely used due to its cost, but has poor latency performance due to the “listen-before-talk” requirement in the ISM bands. Therefore, solutions that combine low cost and high latency performance are needed.

This paper presents a detailed performance evaluation of a 5G DECT NR+ testbed with a focus on latency and data rate. While commercial off-the-shelf (COTS) 5G testbeds are still very expensive, we present a low-cost solution to achieve similar or even better latency performance with NR+. The testbed is composed of two NR+ platforms and will explore different settings such as packet length and Modulation and Coding Scheme (MCS). The main result of the paper is that non-optimized COTS DECT NR+ can provide latency values of 2 ms to 8 ms and a data rate of 2.3 Mbit/s to 2.5 Mbit/s.

Index Terms—DECT-2020 NR, DECT NR+, latency, tactile, robotic

I. INTRODUCTION AND MOTIVATION

In the ever-evolving landscape of telecommunications, the demand increases for advanced technologies capable of supporting emerging applications such as robotics, the metaverse, and the tactile internet. These emerging application scenarios typically require networked delivery of ultra-reliable and low latency (URLLC) sensing and control information. Main technology solutions currently anticipated to meet this challenge include 5G and anticipated successors such as 6G. While these implementations boast impressive capabilities that meet the service-level demands, they commonly face several barriers, such as associated costs for widespread adoption. The DECT-2020 NR+ standard offers a compelling alternative to 5G/6G standards-based implementations with their associated costs.

Delivering the low latency critical for real-time applications at a significantly lower cost, DECT-2020 NR+ represents a viable solution to bridge the gap between technological capability and affordability. For example, the combination of a 5G Core Network and Radio Access Network (RAN) next to required small cells or base stations can readily approach

hundreds of thousands of dollars, not considering the planning, installation, and system integration overheads common to typical campus network deployments. Even smaller deployments thus are cost prohibitive, e.g., for small businesses.

This paper’s main contribution is an overview of the latency performance in different technology environments, focusing on how DECT-2020 NR+ compares to established 5G/6G technologies in a first real-world implementation evaluation. While 5G/6G systems boast high data rates, the DECT-2020 NR+ standard [1] is geared at modest data rates that are achieved with minimal latency — a critical requirement for applications relying on ultra-low latency connectivity. The recent DECT-2020 NR+ standard provides a frame structure similar to cellular technologies, making it suitable for timing guarantees of smaller data payloads. This circumvents CSMA issues that result in limited if any timing guarantees. We provide a detailed analysis of latency and data rate metrics obtained from real-world testing on our dedicated testbed. Through this empirical study, we provide insights that illuminate the practical implications of deploying DECT-2020 NR+ in the context of next-generation telecommunications infrastructure.

II. DECT-2020 NR EVALUATIONS

Digital Enhanced Cordless Telecommunications (DECT) 2020 New Radio (NR) is a radio interface technology which enhances the 5G ecosystem by promising robust, low-latency, and highly reliable wireless communications [1]. This innovative standard operates in the unlicensed spectrum at 1.9 GHz providing a cost-effective and flexible alternative to the licensed 5G spectrum usage. Currently, only a limited number of studies exist that analyze DECT-2020 NR+ employing simulations. Each of them implements its own simulation framework, adding complexity for evaluations. The study in [2] investigates the physical layer performance under industrial conditions. Through their simulations, the authors show that the technology provides low-latency and high-reliability communication, a crucial requirement for real-time industrial applications. In [3], the authors present detailed simulations evaluating the performance of DECT-2020 NR+ for URLLC

and massive Machine-Type Communication (mMTC) applications. This particular study highlights the protocol's ability to achieve low latency and high reliability, again showcasing how DECT-2020 NR+ is suitable for industrial automation and, in the context of the paper, also smart city deployments. In [4], an evaluation of DECT-2020 NR+ is presented that focuses on the multi-hop communication capabilities of the standard. The simulation results reveal that multi-hop configurations enhance coverage and reliability, making the technology ideal for large-scale Internet of Things (IoT) networks in smart cities and rural areas.

While these studies have significantly contributed to the general understanding of the performance attainable with DECT-2020 NR+, they have been limited to individual simulation environments. In contrast, our work represents a pioneering effort in testing real hardware for DECT-2020 NR+. For the first time, we have moved beyond theoretical simulations of individual DECT-2020 NR+ facets to practical, real-world hardware testing. In turn, our contribution provides concrete evidence of the technology's capabilities and performance in actual deployment scenarios.

III. DECT-2020 NR STANDARD OVERVIEW

The European Telecommunications Standards Institute (ETSI) defines the DECT-2020 NR standard in [5]. However, the DECT Forum promotes the standard under DECT NR+ name [6]. Hereafter, we combine both namings and adopt DECT-2020 NR+ throughout this paper.

The architecture overview and all the communication layers are detailed in [1]. In this section, we provide an introduction to some of the main features of the standard.

A. Network Architectures

The DECT-2020 NR+ standard supports the following network architectures:

- 1) Point-to-Point and Point-to-Multipoint: This direct link for low latency communication is suitable to replace cables between, e.g., two isolated industrial devices which require to communicate with each other.
- 2) Single- or multi-cell topologies: One or multiple Radio Devices (RDs) operate in Fixed Termination point (FT) mode and act as base stations.
- 3) Mesh network topology: Each device can communicate to all of its neighbors, and retransmits their messages to extend the range and increase the reliability.
- 4) Internetworking with 3GPP 5G networks: A DECT NR+ network can be connected as untrusted to the 5G Core network via a Non-3GPP Inter-Working-Function (N3IWF) or as trusted via a Trusted Non-3GPP Gateway Function (TNGF).

In this paper, we focus on the Point-to-Point connection as only the physical layer components of the standard were implemented in the hardware we employ for experimentation.

TABLE I
DECT-2020 NR+ MODULATION AND CODING SCHEMES (ACCORDING TO STANDARD VERSION MCS0-MCS4).

MCS Index	Modulation	N_{bps}	R
0	BPSK	1	1/2
1	QPSK	2	1/2
2	QPSK	2	3/4
3	16-QAM	4	1/2
4	16-QAM	4	3/4

B. Physical Layer

The DECT-2020 NR+ physical layer is suitable for operation in frequency bands up to 6 GHz. The standard overall defines 19 different frequency bands from 450 MHz to 5875 MHz. However, in Europe and most of the world, implementations are limited to the unlicensed but regulated DECT frequency band of around 1.9 GHz. Therefore, the hardware that was used for tests supports these 3 bands:

- band 1: 1880 MHz to 1900 MHz
- band 2: 1900 MHz to 1920 MHz
- band 9: 1910 MHz to 1930 MHz

Depending on the subcarrier scaling factor and Fourier transform scaling factor, the transmission bandwidth can vary from 1.728 MHz up to 221.184 MHz.

The typical operation range should theoretically cover several hundreds of meters. However, as described in [7], up to 3 km are achievable within line of sight.

Six different Modulation and Coding Schemes (MCSs) are described in the standard MCS0-MCS5: BPSK, QPSK, 16-QAM, 64-QAM, 256-QAM, and 1024-QAM. Depending on the number of bits per modulation symbol N_{bps} and coding rate R the hardware in the testbed supports the MCS indices provided in Tab. I.

C. Frame Structure

The duration of the radio frame is fixed to $T_{frame} = 10$ ms with 24 slots of $T_{slot} = 0.41667$ ms duration (see Figure 1). These values ensure coexistence with legacy DECT systems as described in [8].

Each DECT-2020 NR+ slot can be divided into 2, 4, 8 or 16 subslots, depending on the configuration. In the tested

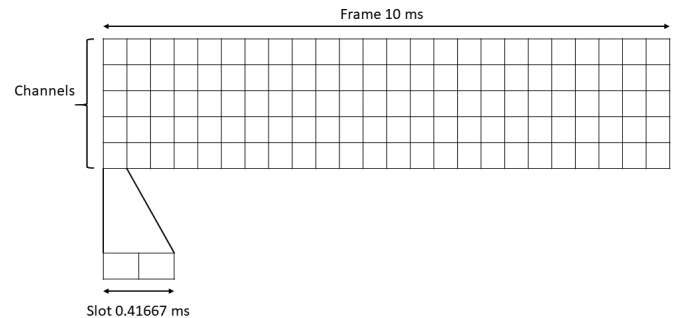


Fig. 1. DECT-2020 NR+ frame structure with two subslots.



Fig. 2. Hardware setup with two Nordic Semiconductor nRF9161 DK development test kits.

hardware, one slot always contains two subslots. For the measurements, we use the number of subslots for the configuration of packet length.

IV. MEASUREMENT SETUP

We employed two development kits from Nordic Semiconductor (nRF9161 DK) for the experimentation and measurements, illustrated in Figure 2. These devices provide access to the implemented DECT-2020 NR+ physical layer through a defined and described Application Programming Interface (API) as part of their device firmware [9].

As illustrated, both kits are connected to the same laptop with a distance of approximately ten cm between each other. The tested hardware was configured to utilize 1.728 MHz transmission bandwidth on channel 1677 with a center frequency of 1899.072 MHz. All measurements employed the maximum transmission power allowed by the hardware implementation (19 dBm).

A. Latency Measurement

The SDK from Nordic Semiconductor provides access to the 64-bit counter that represents modem time. Ahead of each packet's transmission, we generate the packet's payload using a 32-bit sequence number combined with the 64-bit modem time. Subsequently, the minimal required payload size is 12 bytes.

The nRF9161 development kit devices support 5 Modulation and Coding Schemes (MCSs) and allow maximum payload lengths of up to 16 subslots. We measured the latency using all available combinations of MCS and payload lengths described in [9]. However, we excluded configurations that resulted in a payload length of less than 12 bytes.

The broadcast message type was used to measure latency as follows:

- 1) One device configured as transmitter sends the message and switches to reception mode.
- 2) The other device in reception mode receives the message and echos it back, i.e., it copies the payload into the outgoing packet.
- 3) The first device receives the echoed packet.

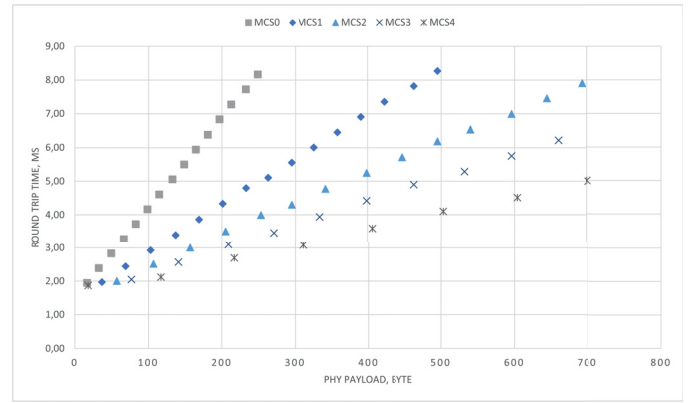


Fig. 3. Round trip time (RTT) averages for nRF9161 with different payloads and MCS settings.

The modem time difference and corresponding round trip time (RTT) are calculated by the receiving device.

B. Throughput Measurement

Throughput measurements were performed with the help of the example application provided by Nordic Semiconductor for reference only. The example application was configured to use MCS4 with 4 slots. The observed maximum data rate for this example setup was between 2.3 Mbit/s and 2.5 Mbit/s. We note that these are only initial values that will be further evaluated in a separate measurement campaign.

V. MEASUREMENT RESULTS

Our measurement results for round-trip-times (RTTs) on actual hardware are described in the following. In Figure 3, we illustrate the RTTs for all configurations supported by the nRF9161 DK with respect to MCSs and payload lengths (except payload length less than 12 bytes). Each point represents the averaged value of 112 measurements.

We initially observe a linear increase in the RTT with an increase in the payload size. This outcome is expected and explained straight-forwardly with the increased transmission times as the payload sizes increase. Similarly, we observe a decrease in the RTT with more modulation steps, i.e., MCS. Again, this behavior is anticipated as increased modulation opportunities result in a higher bit/ baud rate. Finally, we also note that higher payload sizes are attainable for similar RTTs or, alternatively, higher data rates are possible with increased modulation.

We provide a different view in Figure 4 for selected payload lengths given this time in subslots. Comparing the RTTs for the different MCSs but same payload duration in subslots. The RTT depends much more on payload length than on selected MCS. Each additional subslot of payload brings around 0.5 ms for RTT for the same MCS. Increasing or decreasing the MCS-index by one for the same payload length increases or reduces the RTT by around 100 μ s correspondingly. For robotic and tactile applications these values allow to increase robustness by selection of lower MCS-index and longer payload in subslots.

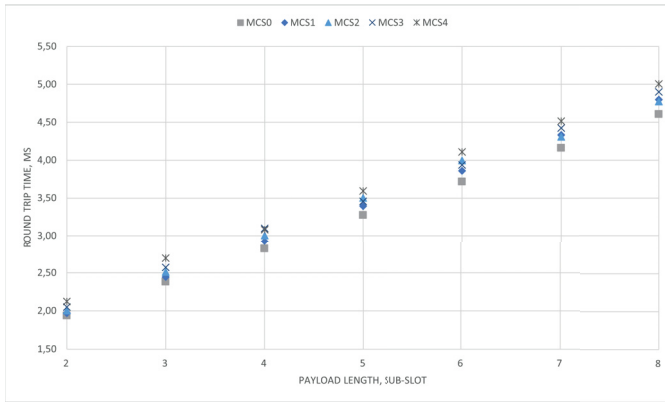


Fig. 4. RTT values obtained for same payload length.

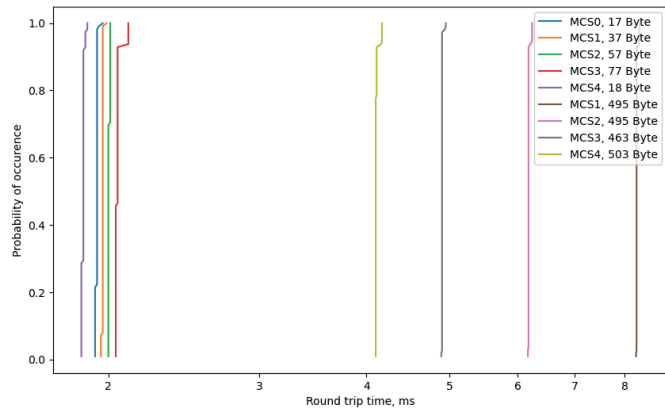


Fig. 5. Cumulative distribution function of RTTs with different combinations of MCS and payload sizes.

In Figure 5, we illustrate the cumulative distribution function of RTTs for different combinations of selected MCSs and payload lengths.

The jitter stays low for all the conducted measurements and RTT values are almost deterministic. This behavior is preferred for robotic applications where constant low latency is crucial for the control loop.

VI. CONCLUSION

While 5G and future 6G deployments provide highly accurate and ultra low latency timing for command and control data required for the tactile internet, including robotics and metaverse applications, this performance comes at a high cost for deployment. Our measurements indicate that latency values as well as packet delay variations, i.e., jitter, are all approaching the thresholds required for tactile internet applications, such as robotics or the metaverse.

Future evaluations will target overall performance measurements including sustainable data rates. Further research will target potentials for further latency reductions and combining DECT-2020 NR+ with other approaches to derive a complete system implementation.

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